Grade 6 Accelerated　　　　Name_Answer Key________
Unit 12　　　　　　　　　　　Date___________________
Lesson 3 Practice　　　　　　Period_________________

Solve each problem using proportional reasoning. Show your work.

For questions 1-2, use the following scenario:

A recipe for salad dressing calls for 3 parts oil for every 1 part vinegar.

1. How many tablespoons (tbsp) of vinegar are needed to make $\frac{1}{2}$ cup of salad dressing? 1 cup = 16 tbsp

$\frac{1 \text{ cup}}{16 \text{ tbsp}} = \frac{\frac{1}{2} \text{ cup}}{8 \text{ tbsp}}$　　oil : vinegar : total　　$\rightarrow 3 : 1 : 4$

$\frac{\text{vinegar}}{\text{total}} \bigg| \; \overset{\times 2}{\frac{1}{4} = \frac{x}{8}} \atop \times 2$　　$x = 2$ tbsp

2. How many tablespoons of oil are needed to make $\frac{1}{2}$ cup of salad dressing?

$\frac{\text{oil}}{\text{total}} \bigg| \; \overset{\times 2}{\frac{3}{4} = \frac{x}{8}} \atop \times 2$　　$x = 6$ tbsp

3. The retail price of grapes is $\$2.17$ per pound. If a customer paid $\$8.68$ for grapes, how many pounds did she buy?

$\frac{\$}{16} \bigg| \; \overset{\times 4}{\frac{2.17}{1} = \frac{8.68}{x}} \atop \times 4$　　$x = 4$ pounds

4. Gregg and Dan are moving boxes. Gregg can move 7 boxes every half hour. Dan can move 10 boxes every half hour. How many hours will it take Gregg and Dan to move all 85 boxes?

$\frac{\text{boxes}}{\text{hours}} \bigg| \; \overset{\times 5}{\frac{17}{0.5} = \frac{85}{x}} \atop \times 5$　　$x = 2.5$ hours

For questions 5-7, use the following scenario:

A recipe for sparkling orange punch calls for $\frac{1}{2}$ gallon of orange sherbet, 6 ounces of frozen orange juice concentrate, and 2 quarts of ginger ale. Mei uses 3 gallons of orange sherbet to make punch for her party.

5. How much ginger ale does Mei need to use for her punch?

$$\frac{\text{sherbert}}{\text{ginger ale}} \quad \Bigg| \quad \frac{\frac{1}{2}\text{ gal}}{2\text{ qt}} \overset{\times 6}{\underset{\times 6}{=}} \frac{3\text{ gal}}{x} \qquad\qquad x = 12 \text{ quarts}$$

6. How much frozen orange juice concentrate does Mei need to use for her punch?

$$\frac{\text{sherbert}}{\text{OJ conc.}} \quad \Bigg| \quad \frac{\frac{1}{2}\text{ gal.}}{6\text{ oz.}} \overset{\times 6}{\underset{\times 6}{=}} \frac{3\text{ gal.}}{x} \qquad\qquad x = 36 \text{ ounces}$$

7. By what factor did Mei scale the recipe up?

6 She made 6 times the original recipe.

For questions 8-10, use the following scenario:

A teacher is planning a field trip to a theme park. For every 6 students, the teacher must have 1 chaperone.

8. What is the ratio of students to total people on the field trip?

$6 : 7$

9. If the teacher purchased 105 total tickets, how many tickets were for students?

$$\frac{\text{students}}{\text{total}} \quad \Bigg| \quad \frac{6}{7} \overset{\times 15}{\underset{\times 15}{=}} \frac{x}{105} \qquad\qquad x = 90 \text{ student tickets}$$

10. If the teacher purchased 63 total tickets, how many tickets were for chaperones

$$\frac{\text{chaperones}}{\text{total}} \quad \Bigg| \quad \frac{1}{7} \overset{\times 9}{\underset{\times 9}{=}} \frac{x}{63} \qquad\qquad x = 9 \text{ chaperone tickets}$$